

Taeyun Yoo

M.S. in Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL

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| [Website](#)

| [LinkedIn](#)

| [Google Scholar](#)

Education

Embry-Riddle Aeronautical University

M.S. in Aviation

Specialization in Aviation Safety Systems

Daytona Beach, FL

Aug 2025–Present

GPA 4.0/4.0

Embry-Riddle Aeronautical University

B.S. in Aeronautical Science

Minor in Air Transportation

Daytona Beach, FL

May 2023

GPA 3.85/4.0

Research Interests

Aviation Safety Analytics; Air Transportation Systems; Trajectory Data Analytics; Terminal and Low-Altitude Airspace Operations

Research Experience

Graduate Researcher

Advisor: Dr. João S. D. Garcia

School of Graduate Studies, College of Aviation

Sep 2025–Present

- ADS-B-based decomposition of approach path stability into sustained and oscillatory deviations
- Wind exposure, approach stability, and go-around initiation in approach and landing operations
- Go-around event-based traffic proximity risk assessment in terminal airspace
- Multi-layer evaluation of AAM disaster response and government-operated public use cases
- ML-based aircraft fuel burn prediction using ADS-B and ACARS data

Graduate Researcher

Advisor: Dr. Ryan Wallace

Center for Advanced Air Mobility

Feb 2026–Present

- LAANC/UASFM-based assessment of shared-airspace exposure between conventional aircraft and UAS operating altitudes
- Nationwide low-altitude traffic density modeling
- ADS-B-based airspace analysis and emerging aviation operations

Publications & Presentations

Publications & Proceedings

[2] Yoo, T., & Garcia, J. S. D. (Under review). Trajectory Analysis of Wind Exposure and Approach Stability in Flight Training Operations.

[1] Yoo, T., & Garcia, J. S. D. (2026). Data-Driven Framework for Assessing Wind Impacts on Approach Stability. *AIAA AVIATION 2026 Forum*. American Institute of Aeronautics and Astronautics.

Research Presentations

INFORMS Annual Meeting (Oral) – Accepted

Institute for Operations Research and the Management Sciences

From Wind Exposure to Go-Arounds: The Mediating Role of Approach Stability in Commercial Aviation

November 2026

San Francisco, CA

UAA 78th Collegiate Aviation Education Conference (Oral) – Accepted

University Aviation Association

September 2026

Denton, TX

The Effect of Wind Exposure on Go-Around Initiation and the Mediating Role of Approach Stability in Flight Operations

AIAA AVIATION 2026 Forum (Oral)

June 2026

American Institute of Aeronautics and Astronautics

San Diego, CA

Data-Driven Framework for Assessing Wind Impacts on Approach Stability

Discovery Day (Poster)

April 2026

Embry-Riddle Aeronautical University

Daytona Beach, FL

Characterizing Air Traffic Density Using Nationwide ADS-B Data

COA Fall Research Colloquium (Oral)

November 2025

College of Aviation, Embry-Riddle Aeronautical University

Daytona Beach, FL

Approach Stability Assessment Using ADS-B Data in Flight Training

Funded Research Projects

2026 Safety Research Seed Grant Program

Boeing Center for Aviation and Aerospace Safety

Graduate Research Lead | PI: João S. D. Garcia | \$15,000

Jul 2026–Jun 2027

An Empirical Analysis of Go-Around and Collision Risk in Support of In-Time Aviation Safety Management Systems

Research Activities Supporting AAM National Strategies

Florida Department of Transportation

Graduate Research Assistant | PI: João S. D. Garcia | \$594,373

Feb–Jun 2026

Advanced Air Mobility in Disaster Response and Other Functions of Government as a Driver for Public Acceptance

Teaching Experience

Graduate Teaching Assistant

Embry-Riddle Aeronautical University

Supervisor: Dr. Dahai Liu

Dec 2025–Present

Teaching support for:

- Statistical Analysis for Aviation/Aerospace
- Research Methods in Aviation/Aerospace
- Graduate Capstone Research Project
- Thesis

Flight Instructor

Aviators Academy

Instructional areas include:

Nov 2023–Jul 2025

- Private Pilot Certificate
- Instrument Rating
- Commercial Pilot Certificate
- Certified Flight Instructor
- Certified Flight Instructor-Instrument
- Flight Review and Instrument Proficiency Check
- Complex Endorsement

Honors & Awards

Best Poster Award, Embry-Riddle Aeronautical University

April 2026

1st Place, College of Aviation Colloquium, Embry-Riddle Aeronautical University

November 2025

Magna Cum Laude, Embry-Riddle Aeronautical University

May 2023

Dean's List (6 semesters), B.S. in Aeronautical Science, Embry-Riddle Aeronautical University

Certifications & Licenses

Commercial Pilot – ASEL, AMEL, Instrument Airplane
Flight Instructor – ASEL, AMEL, Instrument Airplane
Ground Instructor – Advanced, Instrument
Remote Pilot – sUAS

Technical Skills

Python, SPSS, LaTeX

Professional Memberships

Institute for Operations Research and the Management Sciences (INFORMS)
American Institute of Aeronautics and Astronautics (AIAA)
University Aviation Association (UAA)
Royal Aeronautical Society (RAeS)

Leadership & Service

Secretary

Korean Student Association

Embry-Riddle Aeronautical University
2019–2020

Administrative Specialist; Squad Commander

Republic of Korea Military Service

Korea Defense Communication Command
2017–2019

Vice Captain

Table Tennis Club

University of North Dakota
2016–2017